MGU-UGP BSc (HONS) MICROBIOLOGY

IV SEMESTER

MG4DSCMBG200 – ENVIRONMENTAL MICROBIOLOGY

(2024 ONWARDS)

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Time: 1.5 Hours Max. Marks: 50

Part A

Short Notes Questions

Answer any **10** questions

Each question carries **2** marks

1. List any two natural sources of microbial contamination of air. [K]/[CO1]
2. Name two active air sampling devices. [K]/[CO1]
3. Comment on diurnal migration. [U]/[CO2]
4. Define the "Profundal Zone" of a lake. [K]/[CO2]
5. What is phytoremediation? [K]/[CO3]
6. State the primary purpose of using a ‘settle plate’ [U]/[CO1]
7. Name two physical factors that affect microbial growth in natural water. [K]/[CO2]
8. What does a high level of Cladosporium in outdoor air indicate? [K]/[CO1]
9. Write any two uses of biogas. [K]/[CO3]
10. Write any two effects of air pollution on human health. [U]/[CO1]
11. Define spiral length. [K]/[CO2]
12. What is vermicomposting? [K]/[CO3]

(10×2=20)

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

13. Analyse why most marine bacteria are Gram-negative. [An]/[CO2]
14. Explain why *Penicillium* and *Aspergillus* fungi are considered good indicators of indoor air quality and dampness problems. [U]/[CO1]
15. Outline how Upwelling increases ocean productivity. [U]/[CO2]
16. Describe the role of bacteria and fungi in pesticide degradation. [U]/[CO3]

17. Write a short note on the ecological importance of estuaries [U]/[CO2]
18. Discuss the advantages and limitations of bioremediation. [U]/[CO3]

(4×5=20)

Part C

Essay Question

Answer any **1** question

Each question carries **10** marks

19. Discuss the microbial degradation of petroleum hydrocarbons. [U]/[CO3]
20. Explain the zonation of the marine environment with a schematic diagram and describe the characteristics of each zone. [U]/[CO2]

(1×10=10)